Deterministic systems

Transfer operator: gives evolution of densities in $L^1(M)$

Let ρ_t be the density of the state variable at time t.

$$\text{Then, } \mathcal{L}\rho_t := \rho_{t+1} = \frac{\rho_t \circ F^{-1}}{|\det \nabla F| \circ F^{-1}}$$

Koopman operator: L^2 adjoint of transfer operator above: gives evolution of observables on $L^\infty(M)$

$$\mathcal{K} f := f \circ F$$

Note:

$$\langle \mathcal{K} f, \rho \rangle := \int \mathcal{K} f(x) \rho(x)\, dx$$

$$= \int f \circ F(x)\, \rho(x)\, dx =$$

$$\int f(x) \frac{\rho \circ F^{-1}(x)}{|\det \nabla F| \circ F^{-1}(x)}\, dx = \langle f, \mathcal{L}\rho \rangle$$

Koopman operator approximation

Extended Dynamic mode decomposition

$A \in \mathbb{C}^{b \times m}$ with elements $A_{ki} = \phi_k(x_i)$

$B \in \mathbb{C}^{b \times m}$ with element $B_{ki} = \mathcal{K}\phi_k(x_i)$

$= \phi_k(F(x_i))$

$\Phi := \{\phi_1, \phi_2, \dots, \phi_b\}$: some set of basis functions

Suppose $\mathcal{K}\phi_l = \sum_{k=1}^{b} c_k^l \phi_k + r$

r: residual function

Then,

$$\langle \mathcal{K}\phi_l, \phi_{l'} \rangle = \sum_{k=1}^{b} c_k^l \langle \phi_k, \phi_{l'} \rangle + \langle r, \phi_{l'} \rangle$$

If Φ is an orthonormal basis

$$\langle \mathcal{K}\phi_l, \phi_{l'} \rangle = c_{l'}^l + \langle r, \phi_{l'} \rangle$$

Galerkin projection idea: set $\langle r, \phi_{l'} \rangle = 0$.
for any $l' \in [b]$.

$\Rightarrow \langle \mathcal{K}\phi_l, \phi_{l'} \rangle := c_{l'}^{l}$

Want to find matrix K
with entries $c_l^{l'}$

This matrix represents action of
$\mathcal{K}$ on Φ.

Found by minimizing r at
points $x_1, x_2, \ldots, x_m$.

That is, $K := \min_U \| UA - B \|^2$

then $K \in \mathbb{C}^{b \times b}$ approximately

has elements $K_{kl} = c_l^k$

$\approx$ "Coeff of $\mathcal{K}\phi_k$ along ϕ_l".

Closed form solution $K = BA^{\dagger}$

j^{th} row of B

$$\|u_j^T A - B_j^T\|^2 \Rightarrow \|A^T u_j - B_j\|^2$$

A^T : tall and thin matrix
overdetermined least squares

$$u_j = (A A^T)^{-1} A B_j$$

$$\Rightarrow u_j^T = B_j^T A^T (A A^T)^{-1}$$

$$\Rightarrow K = B A^T (A A^T)^{-1}$$

(stack u_j^T)

assuming A is full row rank

T = conjugate transpose

Now, $$(A A^T)_{kl} = \sum_{i=1}^{m} \phi_k(x_i)\, \phi_l^*(x_i)$$

$$(B A^T)_{kl} = \sum_{i=1}^{m} \phi_k(Fx_i)\, \phi_l^*(x_i)$$

$\Rightarrow$ as $m \to \infty$, $$\frac{1}{m}(A A^T)_{kl} \to \langle \phi_k, \phi_l \rangle_\mu$$

and $\frac{1}{m}(BA^T)^{\phi}_{k\ell} \Rightarrow \langle \mathcal{K}\phi_k, \phi_\ell \rangle$

$$\Rightarrow K \longrightarrow \left[\langle \mathcal{K}\phi_k, \phi_\ell \rangle_\mu\right]_{k,\ell} \left[\langle \phi_k, \phi_\ell \rangle_\mu\right]^{-1}_{k,\ell}$$

Right eigenvectors are called "Koopman modes" and left eigenvectors EDMD modes

$$v^T K = \lambda v^T \quad \text{(left eigenvector)}$$

Then, $\mathcal{K} v^T \underset{\swarrow}{\underline{\Phi}}(x) = v^T \underline{\Phi}(F(x))$

$(\phi_1(x), \dots, \phi_b(x))^T$

$$= v^T K \underline{\Phi}(x)$$

$$= \lambda v^T \underline{\Phi}(x)$$

(as $m \to \infty$)

$\Rightarrow$ $v^T \underline{\Phi}(x)$ are eigenfunctions of $\mathcal{K}$.